The Andrew and Erna Viterbi Faculty of
ELECTRICAL & COMPUTER ENGINEERING

TECHNION
Israel Institute of Technology

Final Presentation

Quantum Enhanced Electron Microscopy

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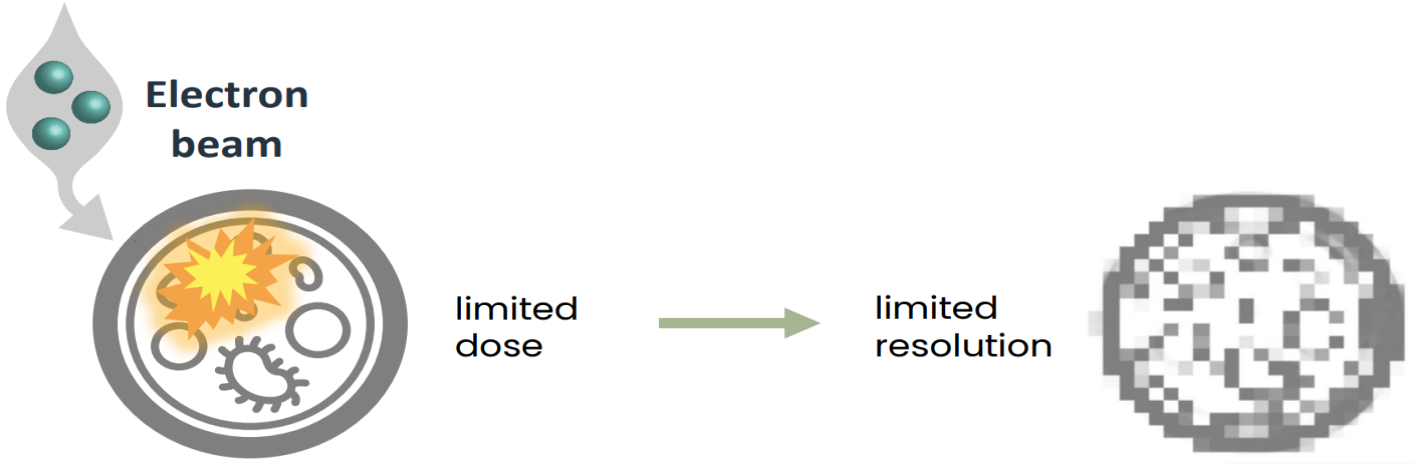
Outline

- Motivation & Goal
- Background: Quantum Metrology & QFI
- Literature Survey: Spin Squeezing and single-parameter limitations
- Chosen Solution: Multipixel phase estimation
- Results: State optimization under mixed noise
- Conclusions & Future Work

Image Precision vs. Radiation Damage

- Efficient microscopy is a critical requirement across various scientific fields
- Transmission Electron Microscopes:
  - + Achieve atomic resolution
  - - Highly limited by beam-induced damage



Our Goal: Maximize Resolution Under A Finite Dose

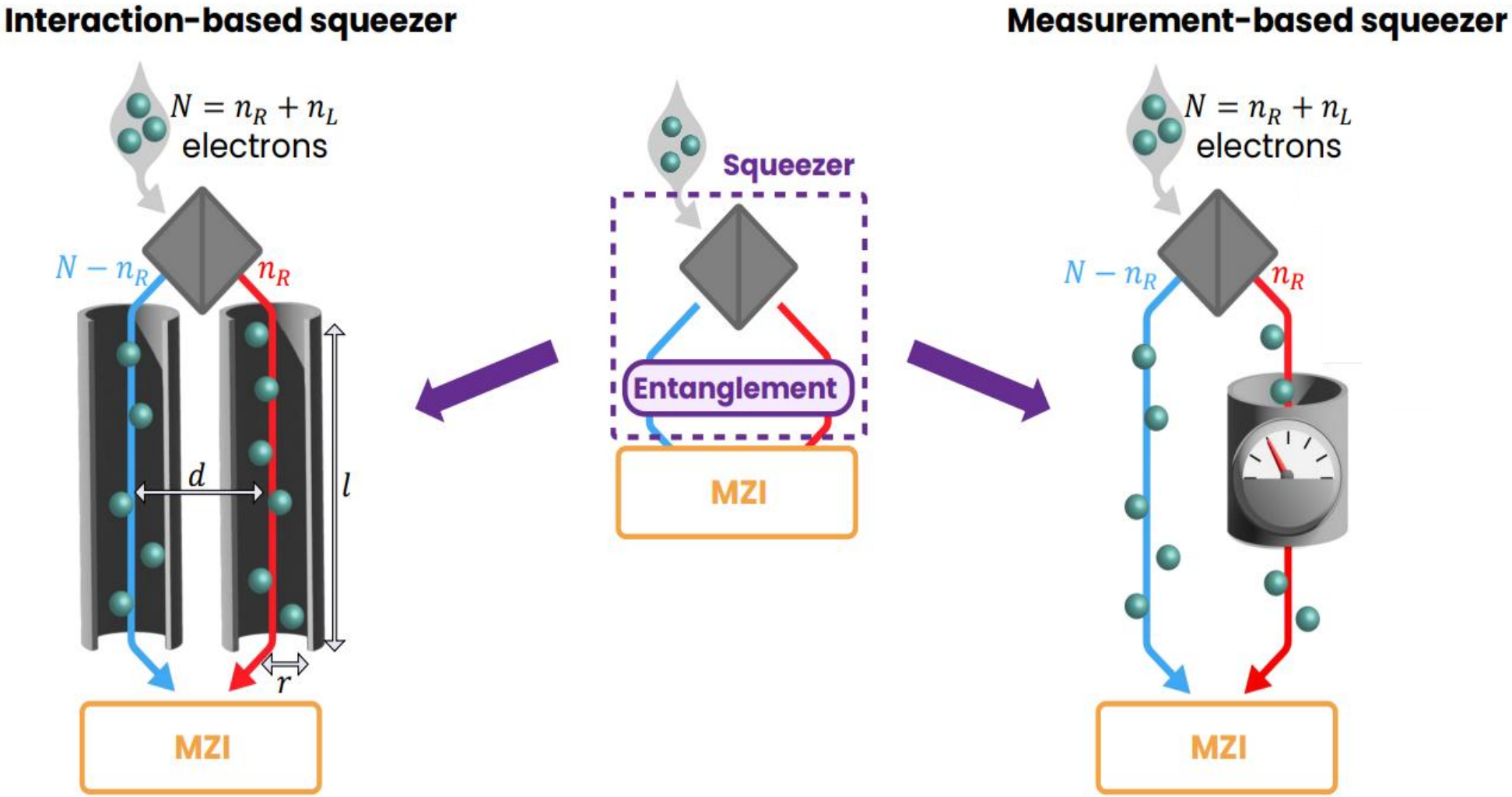
- Maximally utilize information for a given dose
  - Classical Limit (SQL): $\Delta\phi \geq \frac{1}{\sqrt{N}}$
  - Heisenberg Limit: $|\Delta\phi| \geq \frac{1}{N}$

The Method

- Explore techniques for quantum enhanced EM
- Learn about current theoretical solutions
- Implement a realistic noisy setup
- Analyze the optimized results

Literature Survey: Spin Squeezing Techniques

- Rec
- Two



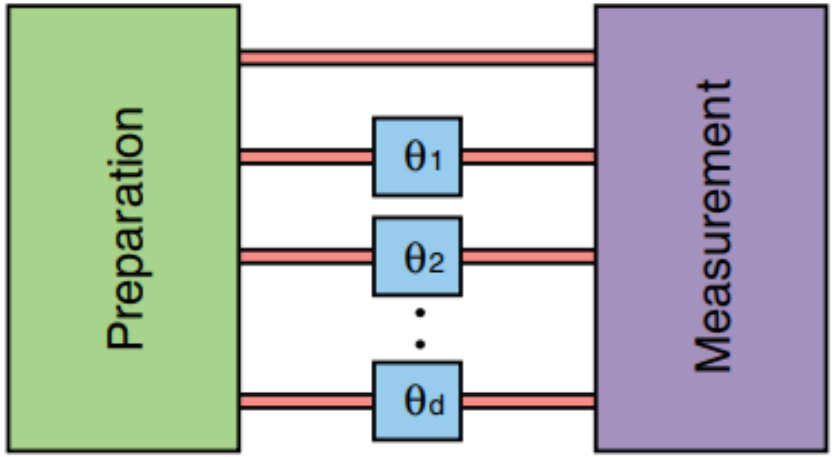
So, is Spin Squeezing optimal for our goal?

+ High robustness against electron loss
+ Feasible with existing microscope tech
- **Optimized for single-parameter phase estimation**

Chosen Solution: Multipixel Phase Estimation

- **Imaging is inherently a multiparameter estimation problem**
- Preparation: N particles distributed across $d+1$ modes
- Interaction: Signal modes acquire d independent phases
- Measurement: Maps accumulated phases to an image

- For pure states: $\left|\Delta\theta_{opt}\right|^2 = \frac{\left(1+\sqrt{d}\right)^2 d}{4N^2}$ [1]
- $\mathcal{O}(d)$ advantage over d separate estimation



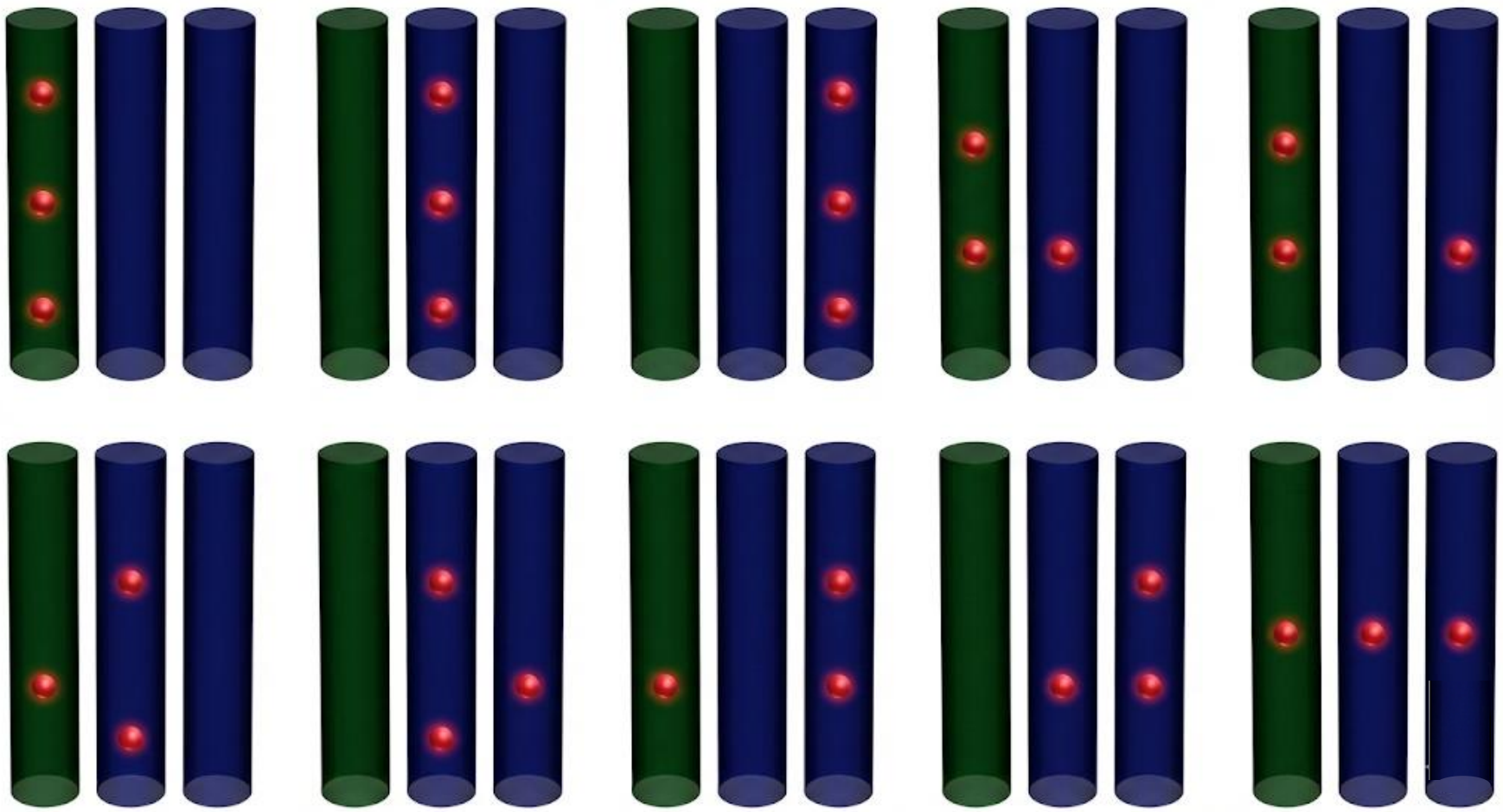
Multipixel Phase Estimation - Preparation

- A general pure input state

$$|\psi\rangle = \sum_{l=1}^{D} \alpha_l |N_{l,0}, \dots N_{l,d}\rangle \equiv \sum_{l=1}^{D} \alpha_l |\boldsymbol{N_l}\rangle$$

- Distributes N particles across $d+1$ modes
- $D = \frac{(N+d)!}{N!d!}$ configurations

Example for $N = 3, d = 2 \Longrightarrow D = 10$



Multipixel Phase Estimation – Interaction

- Sample acts as a multi-mode phase shifter
- Each mode applies the unitary transformation $U_{\boldsymbol{\theta}} = e^{i\sum_{m=0}^{d} \hat{N}_m \theta_m}$
- $\boldsymbol{\theta} = \{\theta_0, \theta_1, \dots, \theta_d\}$ are the phases accrued by the modes
- The evolved state:

$$|\psi_{\boldsymbol{\theta}}\rangle = U_{\boldsymbol{\theta}}|\psi\rangle = \sum_{l}^{D} \alpha_l e^{i\boldsymbol{N}_l \cdot \boldsymbol{\theta}} |\boldsymbol{N}_{\boldsymbol{l}}\rangle$$

Multipixel Phase Estimation – Measurement

- QFI matrix I_θ quantifies available phase information
- Precision bounded by Cramér-Rao Bound: $Cov(\boldsymbol{\theta}) \geq (MI_{\boldsymbol{\theta}})^{-1}$
- Taking the trace (with $M = 1$): $|\Delta\boldsymbol{\theta}| \equiv Tr[Cov(\boldsymbol{\theta})] \geq Tr\left[I_{\boldsymbol{\theta}}^{-1}\right]$
- I_θ for **pure** states [2]

$$I_\theta = 4\sum_i |\alpha_i|^2 \boldsymbol{N}_i \boldsymbol{N}_i^T - 4\sum_{i,j} |\alpha_i|^2 |\alpha_j|^2 \boldsymbol{N}_i \boldsymbol{N}_j^T$$

- Solution: A $N00N$ state [1]:
  $|\psi\rangle_{opt} = \beta|N, 0, \dots 0\rangle + \alpha(|0, N, \dots, 0\rangle + \cdots + |0,0, \dots N\rangle)$

Reality is Noisy

- The described method assumes noiseless probes
- Real microscopes suffer from loss
- This implies a possible different solution than the $N00N$ state
- **Our work: Optimizing for mixed-noise channels**

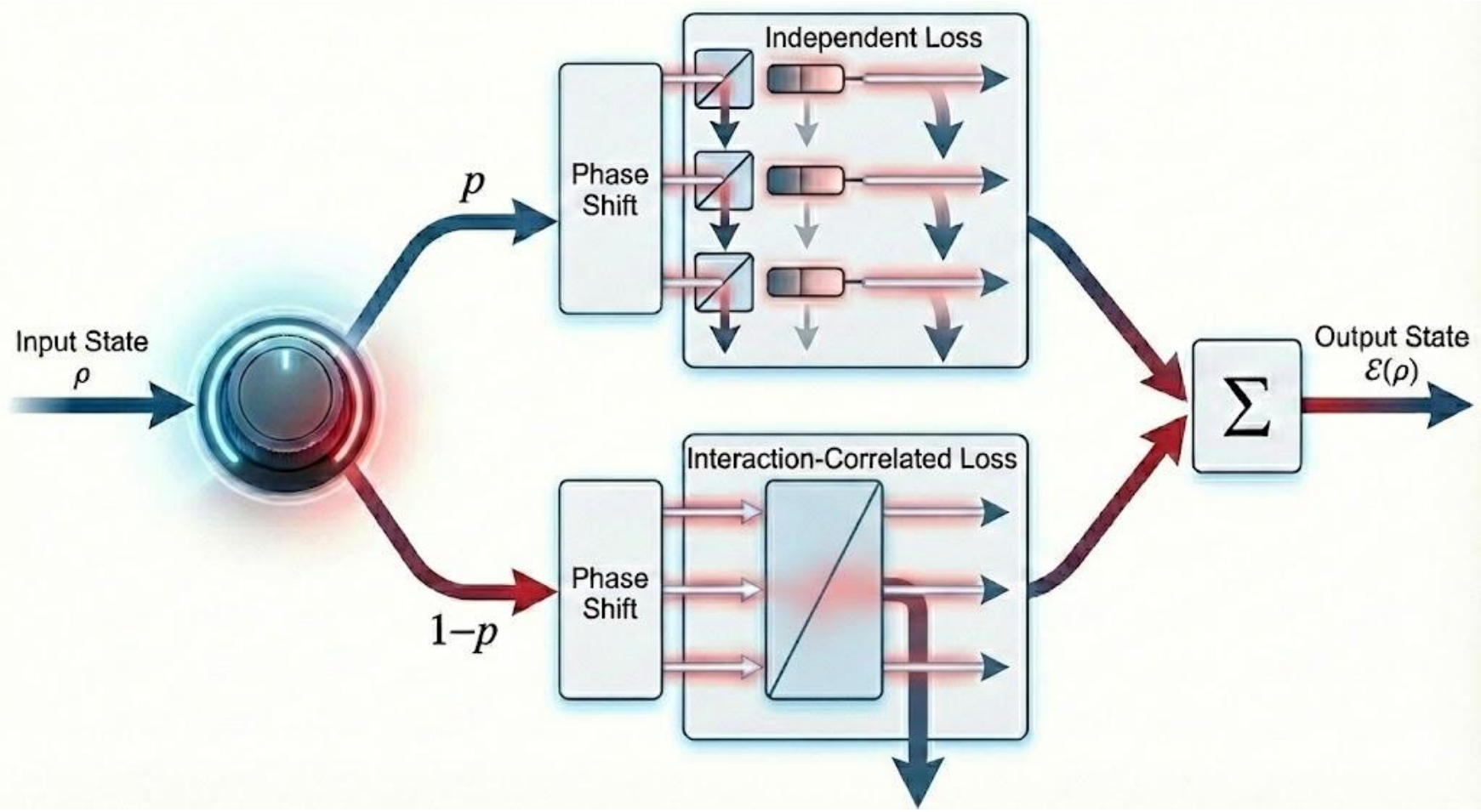
Implementation: Realistic Noise Modeling

- Model: Particle loss via beam splitter interaction
- Transmission parameter η scales from 0 to 1
- The global channel transformation:

$$\mathcal{E}(\rho) = p\mathcal{E}_{ind}(\rho) + (1-p)\mathcal{E}_{int}(\rho)$$

- $\mathcal{E}_{ind}$: Independent particle loss
- $\mathcal{E}_{int}$: Interaction-correlated loss
- Parameter p balances noise source distribution

Realistic Noise Modeling: Illustration



Numerical Implementation & Workflow

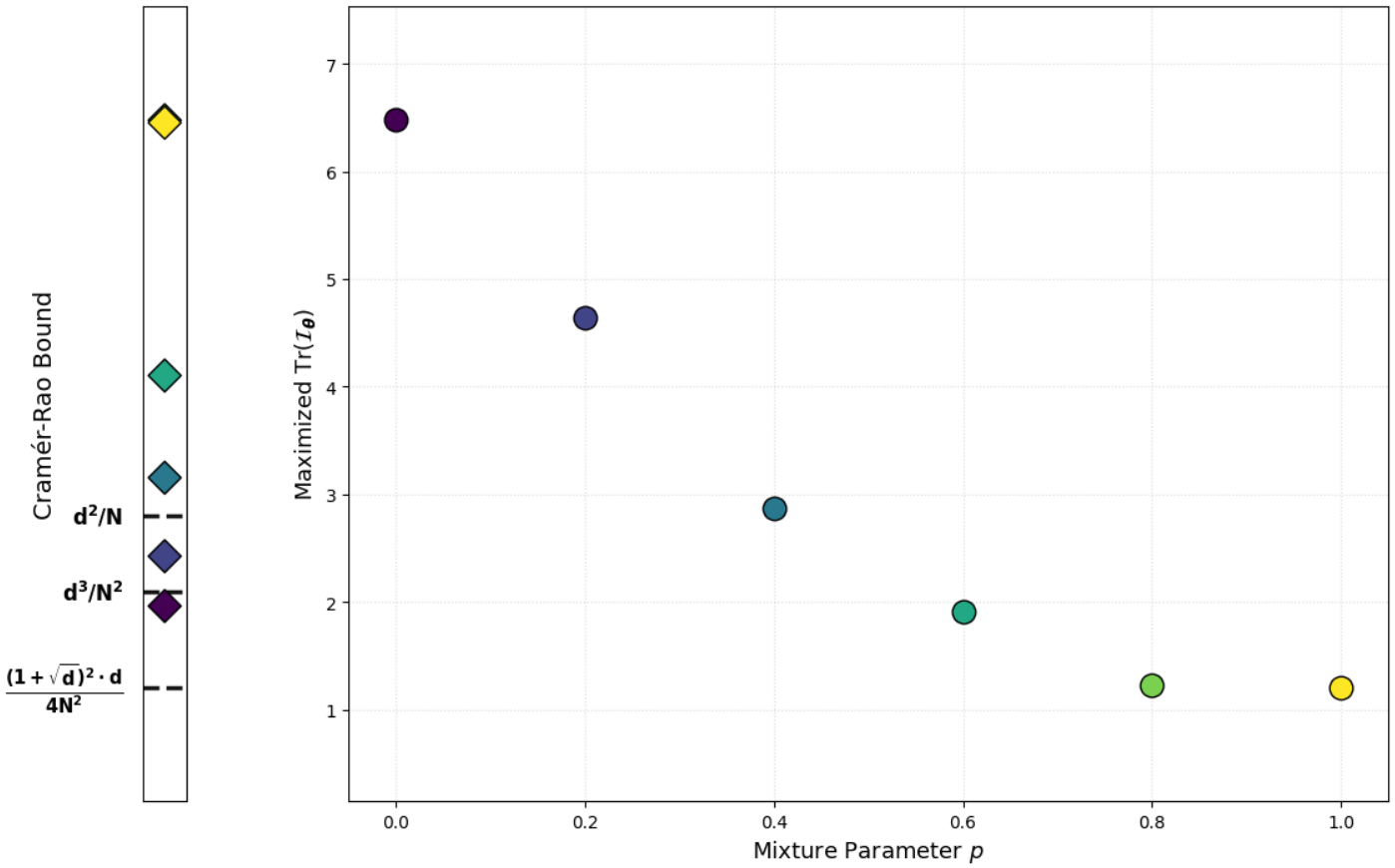
- Basis: Fock space of D configurations
- $\rho_{out} = p \cdot \mathcal{E}_{ind}\left(U_{\boldsymbol{\theta}}\rho_{in}U_{\boldsymbol{\theta}}^{\dagger}\right) + (1-p) \cdot \mathcal{E}_{int}\left(U_{\boldsymbol{\theta}}\rho_{in}U_{\boldsymbol{\theta}}^{\dagger}\right)$
- I_θ for **non pure** states [2]

$$I_{\boldsymbol{\theta}}[a,b] = \sum_{i,j=0,\lambda_i+\lambda_j\neq 0}^{d-1} \frac{2Re\left(\langle\lambda_i|\partial_a\rho|\lambda_j\rangle\langle\lambda_j|\partial_b\rho|\lambda_i\rangle\right)}{\lambda_i+\lambda_j}$$

- Optimal coefficients of ρ_{in}

$$\min_{\boldsymbol{\alpha}\in\mathbb{C}^D:\|\boldsymbol{\alpha}\|=1} -Tr[I_{\boldsymbol{\theta}}(\boldsymbol{\alpha},\eta,p,\boldsymbol{\theta})]$$

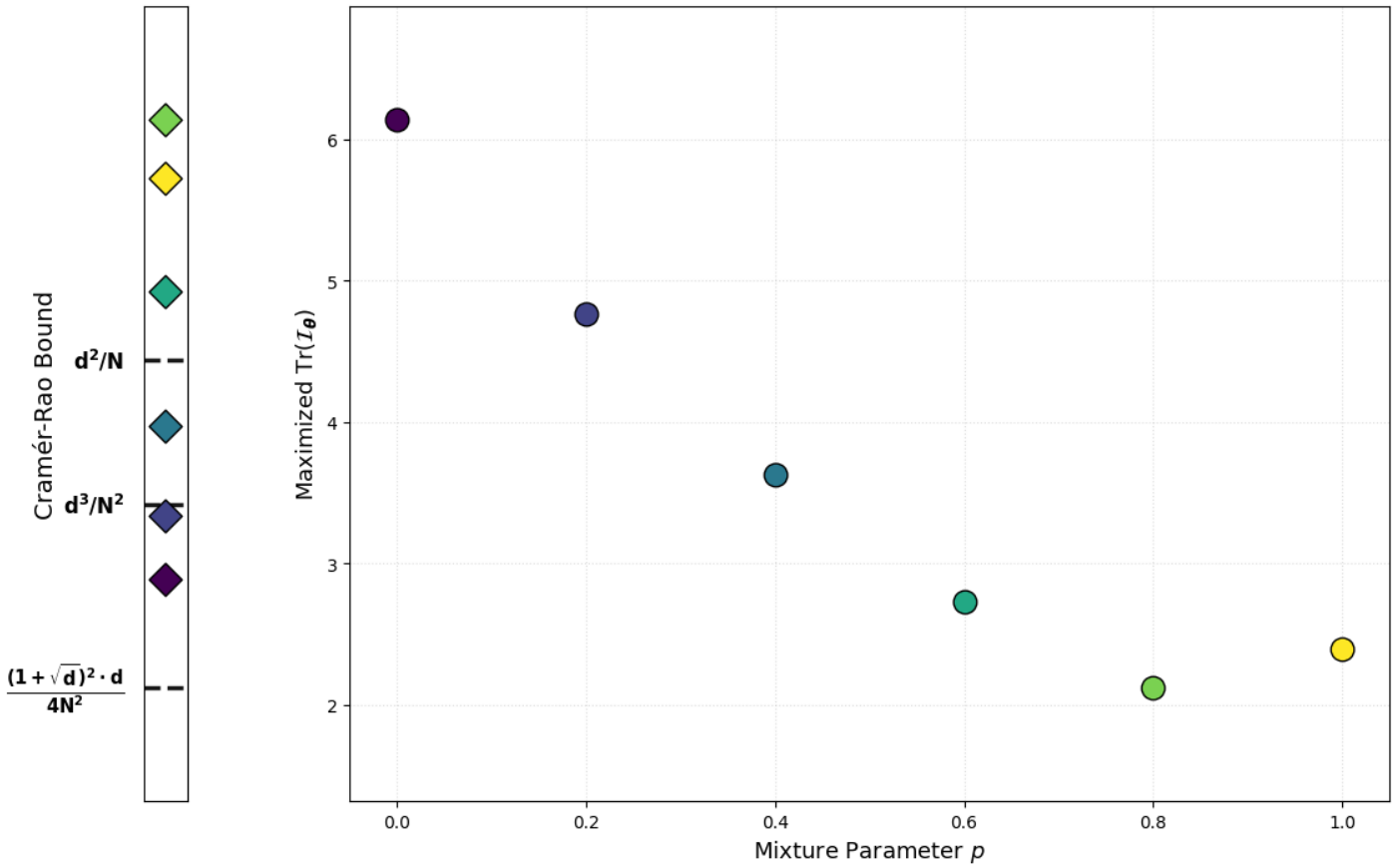
Results - $N = 3, d = 2, \eta = 0.2$



Optimized States $|\psi\rangle$

- p=0.00: $0.51|0,0\rangle + 0.02|1,2\rangle + 0.86|3,0\rangle$
- p=0.20: $0.53|0,0\rangle + 0.05|1,2\rangle + 0.85|3,0\rangle$
- p=0.40: $0.44|0,0\rangle + 0.05|0,2\rangle + 0.44|1,0\rangle + 0.06|1,2\rangle + 0.26|2,0\rangle + 0.74|3,0\rangle$
- p=0.60: $0.37|0,0\rangle + 0.07|0,2\rangle + 0.53|1,0\rangle + 0.11|1,2\rangle + 0.32|2,0\rangle + 0.68|3,0\rangle$
- p=0.80: $0.25|0,0\rangle + 0.09|0,2\rangle + 0.60|1,0\rangle + 0.19|1,2\rangle + 0.32|2,0\rangle + 0.66|3,0\rangle$
- p=1.00: $0.25|0,0\rangle + 0.35|0,1\rangle + 0.33|0,2\rangle + 0.22|0,3\rangle + 0.35|1,0\rangle + 0.42|1,1\rangle + 0.32|1,2\rangle + 0.33|2,0\rangle + 0.32|2,1\rangle + 0.22|3,0\rangle$

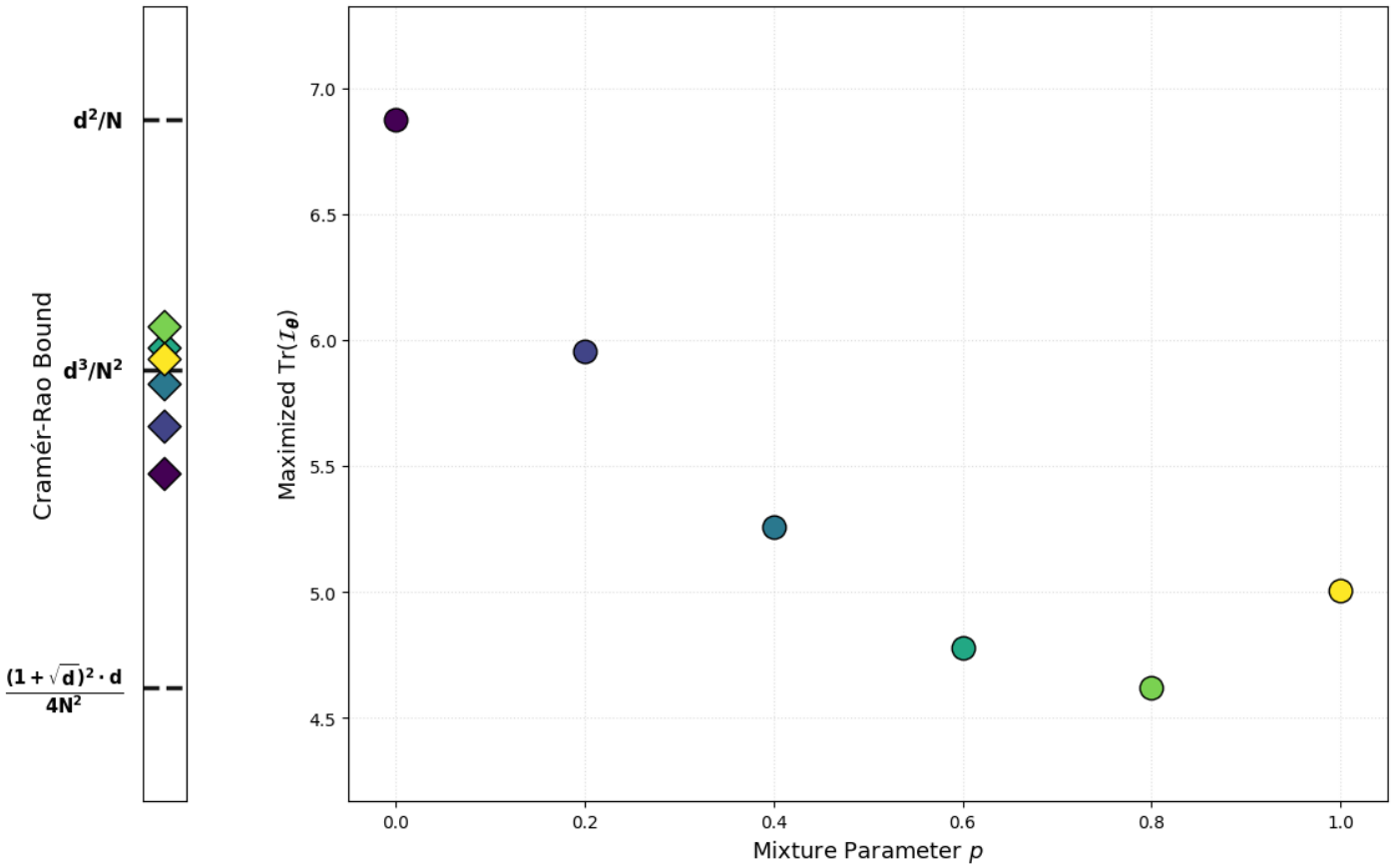
Results - $N = 3, d = 2, \eta = 0.4$



Optimized States $|\psi\rangle$

- p=0.00: $0.55|0,0\rangle + 0.17|1,2\rangle + 0.82|3,0\rangle$
- p=0.20: $0.55|0,0\rangle + 0.23|1,2\rangle + 0.80|3,0\rangle$
- p=0.40: $0.55|0,0\rangle + 0.33|1,2\rangle + 0.77|3,0\rangle$
- p=0.60: $0.54|0,0\rangle + 0.12|0,2\rangle + 0.14|1,0\rangle + 0.43|1,2\rangle + 0.20|2,0\rangle + 0.67|3,0\rangle$
- p=0.80: $0.42|0,0\rangle + 0.38|0,1\rangle + 0.29|0,2\rangle + 0.40|0,3\rangle + 0.26|1,0\rangle + 0.39|2,0\rangle + 0.26|2,1\rangle + 0.38|3,0\rangle$
- p=1.00: $0.44|0,0\rangle + 0.30|0,1\rangle + 0.26|0,2\rangle + 0.30|0,3\rangle + 0.30|1,0\rangle + 0.27|1,1\rangle + 0.34|1,2\rangle + 0.26|2,0\rangle + 0.34|2,1\rangle + 0.30|3,0\rangle$

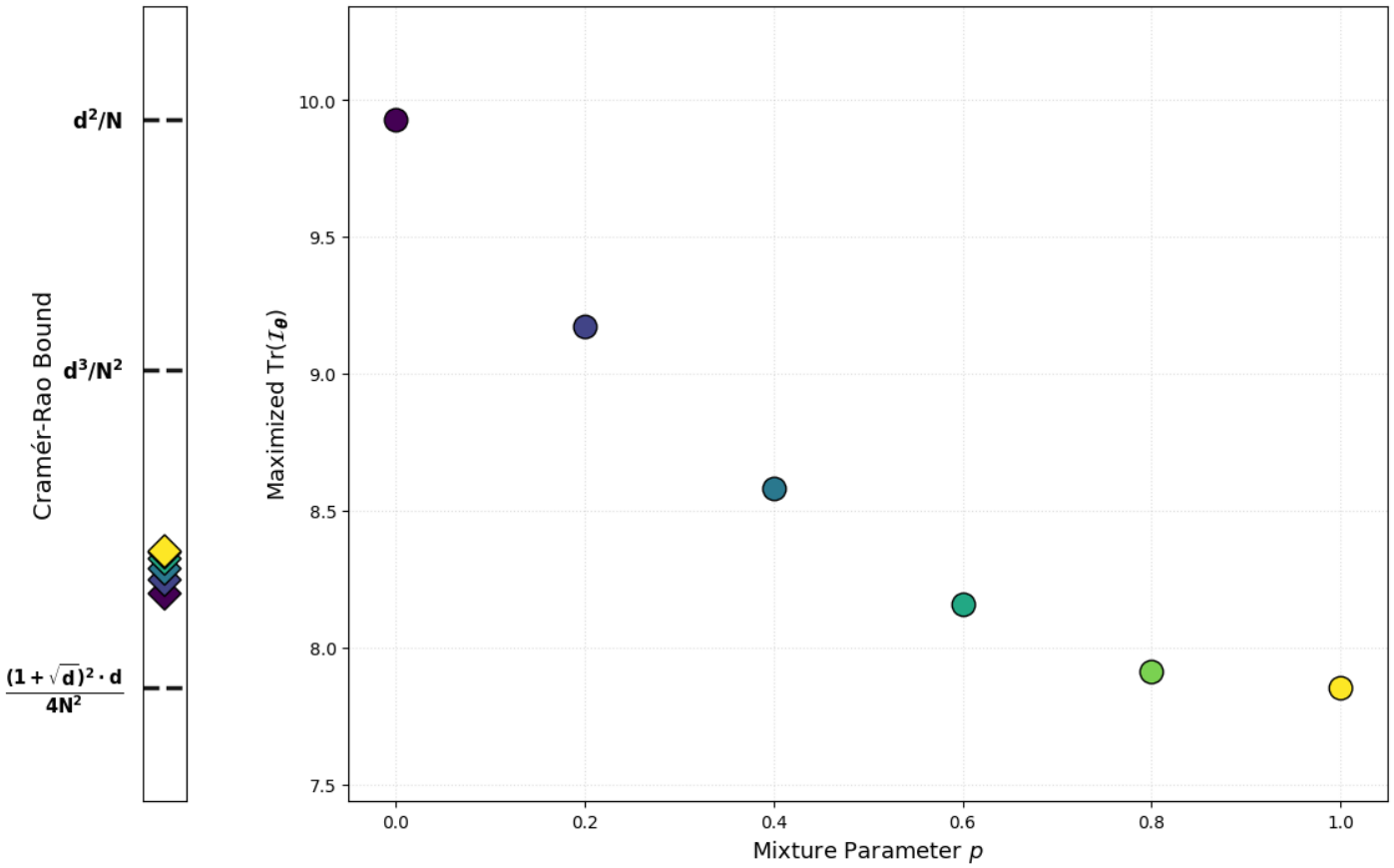
Results - $N = 3, d = 2, \eta = 0.6$



Optimized States $|\psi\rangle$

- p=0.00: $0.55|0,0\rangle + 0.54|0,3\rangle + 0.64|3,0\rangle$
- p=0.20: $0.55|0,0\rangle + 0.57|0,3\rangle + 0.61|3,0\rangle$
- p=0.40: $0.54|0,0\rangle + 0.59|0,3\rangle + 0.60|3,0\rangle$
- p=0.60: $0.53|0,0\rangle + 0.60|0,3\rangle + 0.60|3,0\rangle$
- p=0.80: $0.53|0,0\rangle + 0.59|0,3\rangle + 0.20|2,1\rangle + 0.58|3,0\rangle$
- p=1.00: $0.53|0,0\rangle + 0.48|0,3\rangle + 0.36|1,2\rangle + 0.36|2,1\rangle + 0.48|3,0\rangle$

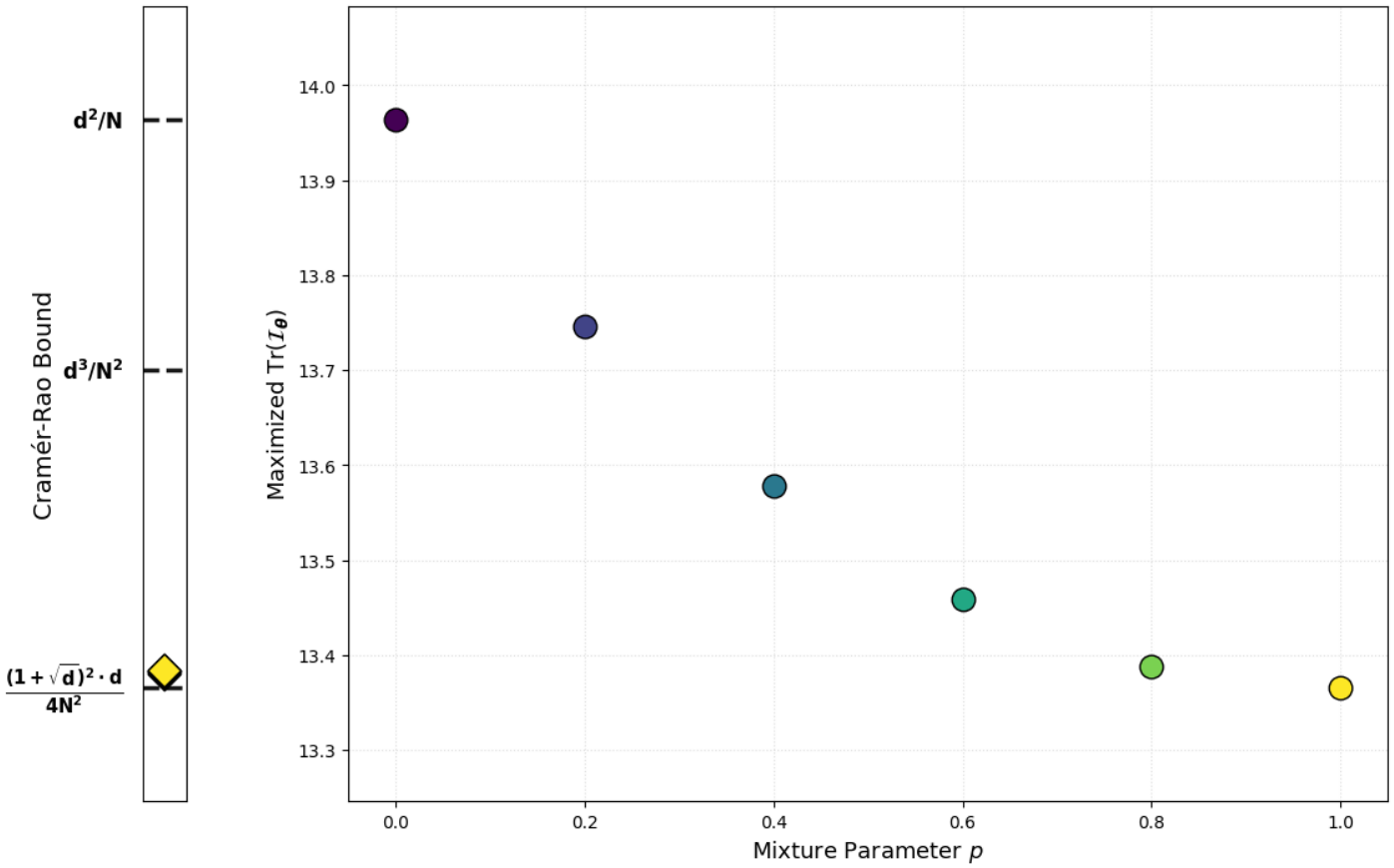
Results - $N = 3, d = 2, \eta = 0.75$



Optimized States $|\psi\rangle$

- p=0.00: $0.59|0,0\rangle + 0.57|0,3\rangle + 0.58|3,0\rangle$
- p=0.20: $0.58|0,0\rangle + 0.57|0,3\rangle + 0.58|3,0\rangle$
- p=0.40: $0.57|0,0\rangle + 0.58|0,3\rangle + 0.58|3,0\rangle$
- p=0.60: $0.57|0,0\rangle + 0.58|0,3\rangle + 0.58|3,0\rangle$
- p=0.80: $0.57|0,0\rangle + 0.58|0,3\rangle + 0.58|3,0\rangle$
- p=1.00: $0.56|0,0\rangle + 0.57|0,3\rangle + 0.12|1,2\rangle + 0.12|2,1\rangle + 0.57|3,0\rangle$

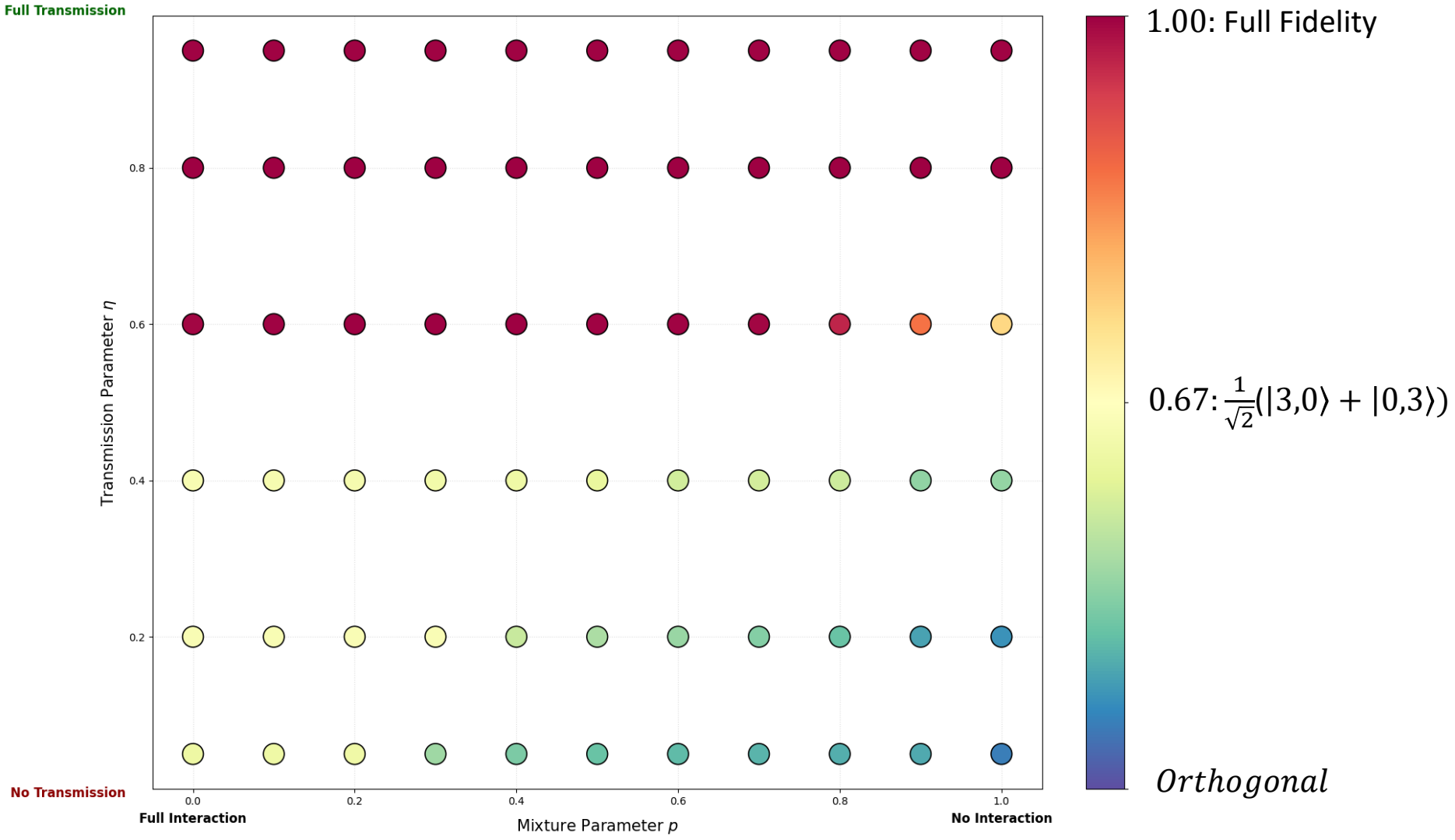
Results - $N = 3, d = 2, \eta = 0.95$



Optimized States $|\psi\rangle$

- p=0.00: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$
- p=0.20: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$
- p=0.40: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$
- p=0.60: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$
- p=0.80: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$
- p=1.00: $0.63|0,0\rangle + 0.55|0,3\rangle + 0.55|3,0\rangle$

Fidelity with $|\psi\rangle = \frac{1}{\sqrt{3}}(|0,0\rangle + |3,0\rangle + |0,3\rangle)$



Conclusions

- Multipixel estimation dramatically improves resolution for a given dose
- For $\eta \geq 0.6$, $|\psi\rangle \approx \frac{1}{\sqrt{3}}(|0,0\rangle + |3,0\rangle + |0,3\rangle)$ is optimal
- Quantum Advantage: Precision beats SQL even at 80% loss

Future Work

- Scaling: Increase N and d using GPU-accelerated computing
- Explore methods to prepare optimal W-states
- Model validation against real scattering data

References

1. P. C. Humphreys, M. Barbieri, A. Datta, and I. A. Walmsley, "Quantum Enhanced Multiple Phase Estimation," *Phys. Rev. Lett.*, vol. 111, no. 7, p. 070403, Aug. 2013.
2. J. Liu, H. Yuan, X. Lu, and X. Wang, "Quantum Fisher information matrix and multiparameter estimation," *J. Phys. A: Math. Theor.*, vol. 53, no. 2, p. 023001, Jan. 2020, doi: 10.1088/1751-8121/ab5d4d.